

Assignment 6

Bronze Tier (Grade 5) – Deadline: Week 10

Objective:

Build a relational database infrastructure that fully supports CRUD operations, filtering, and sorting between related entities.

Requirements:

- **Database Relationships:**
 - Design at least two classes in a **1-to-many** or **many-to-many** relationship.
- **Full CRUD Support:**
 - Implement Create, Read, Update, and Delete operations for all entities.
- **Filtering and Sorting:**
 - Enable users to filter and sort entity data based on multiple criteria.
- **Preferably Use ORMs:**
 - Use an ORM like Sequelize, TypeORM, Hibernate, or Entity Framework to manage database operations efficiently.

Reward:

Completing the Bronze Tier earns you a grade of **5**.

Silver Tier (Grade 7) – Deadline: Week 10

Objective:

Stress-test your database by populating it with huge volumes of data and optimize performance for complex queries.

Requirements:

- **Populate Tables:**

- Use libraries like **Faker** to generate and insert **more than 100,000 entries** into each related table.
- **Performance Optimization:**
 - Optimize data fetching times for at least one **complex statistical query** involving your related entities.
 - Implement database optimizations such as **indices, query optimization techniques**, or other performance enhancements.
- **Performance Testing:**
 - Use **JMeter** to simulate multiple requests and illustrate how your optimized endpoint performs under load.

Reward:

Completing the Silver Tier earns you 2p.

 Gold Tier (Grade 10) – Deadline: Week 10

Objective:

Implement user monitoring infrastructure, supporting user authentication, user roles, and behavior tracking through backend logging and analysis.

Requirements:

- **Authentication System:**
 - Implement **basic register/login functionality** (no OAuth needed yet).
 - Each user must have their own separate entities.
- **User Roles:**
 - Define at least **two roles**:
 - **Regular User**
 - **Admin**
- **Logging System:**

- Track every CRUD operation in a **log table**, storing:
 - Which user performed the action.
 - What action was performed.
 - When it was performed.
- **Background Monitoring Thread:**
 - A separate backend thread should:
 - Periodically analyze the log table.
 - Detect unusual activity (e.g., CRUD actions performed at a suspiciously high frequency within a given time window).
 - Add suspicious users to a **monitored users list**.
- **Admin Dashboard:**
 - Only admins can view the **list of monitored users** on the frontend.
- **Simulated Attack Scenario:**
 - Simulate a scenario where one user performs a high number of CRUD operations in a short time.
 - Show how the backend thread detects this and adds the user to the monitored users list.
 - Ensure the admin can view this list once logged in.

Reward:

Completing the Gold Tier earns you 3p.